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Organisation: Khumo Advertising Billboards (Vryburg)

Industry :Media

Addressing block : 192.168.56.0/24

CMPG325 – Milestone 1 – Client Design Review

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1. Client requirements

1.1 Requirements specified in the project brief

- Assigned organisation: Khumo Advertising Billboards (Vryburg)
- Industry: Media
- Assigned addressing block: 192.168.56.0/24
- Client change request (CR15): a second internet connection is added for resilience and must be integrated.
- Design constraint: VoIP traffic must be separated from data traffic.
- Assigned networking challenge: Wireless Security (WPA2-PSK hardening; Enterprise optional extension) to be configured, verified, and demonstrated within the client network.
- Network design requirements: implement in Cisco Packet Tracer; design an appropriate topology; configure all necessary routers, switches, and end devices; demonstrate successful connectivity and testing.

1.2 Engineering assumptions

The project brief does not specify organisational detail such as departments, user counts, or device placement. Where this information is not provided, the following network-engineering assumptions have been made, each justified against the assigned scenario. These assumptions form the baseline for the topology, VLAN design, and IP addressing plan that follow.

Departments

The brief identifies the client as a Media-industry organisation. Three logical departments are assumed to reflect a realistic small billboard advertising business: Management/Admin (administration and client liaison), Creative/Design (billboard content design), and Production/Logistics (installation and maintenance coordination).

Users and devices

An SMB environment of approximately 30 users is assumed, distributed across desktop workstations, IP phones, and wireless-capable devices. This reflects a single-site office rather than a multi-branch enterprise, consistent with the brief not indicating additional locations.

Wireless

Wireless access is assumed to be required for the Creative/Design department and shared meeting rooms, served by a dedicated Wireless Access Point. This gives a concrete, defensible location for the assigned WPA2-PSK hardening challenge and reflects the mobility needs of design staff moving between workstations and meeting spaces.

VoIP

To satisfy the design constraint separating VoIP traffic from data traffic, IP phones are assumed to be deployed in the Management/Admin department only, on a dedicated Voice VLAN. Management/Admin is the natural point of client contact for billboard bookings and enquiries, so voice traffic is concentrated there rather than distributed across every department. Creative/Design and Production/Logistics are assumed to rely on data

services only, keeping the voice footprint small and clearly scoped, which is appropriate for a business of this size.

Redundancy (CR15)

A dual-homed ISP setup two independent edge routers, each terminating a separate internet connection is assumed to satisfy the change request for resilience. This allows the network to fail over automatically if the primary connection is lost, keeping cloud-based scheduling, design, and communication tools available.

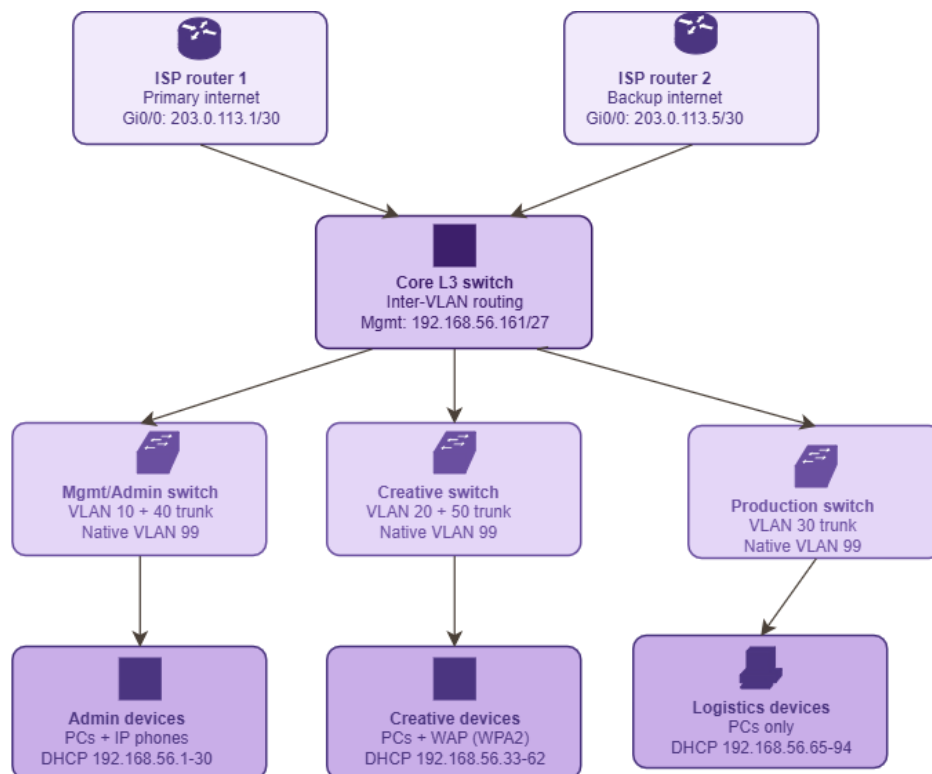
2. Physical topology

2.1 Design explanation

The physical topology follows a three-tier hierarchical model (Edge, Core, Access), chosen to satisfy CR15's resilience requirement and to keep departmental VLANs cleanly separated at the access layer.

- **Edge/ISP Layer:** Two independent ISP routers (**ISP 1** and **ISP 2**) provide the redundant internet connectivity required by CR15. If either link fails, the network continues operating on the remaining connection.
- **Core Layer:** A single Cisco multilayer (Layer 3) switch acts as the central gateway for every VLAN and performs all inter-VLAN routing via Switch Virtual Interfaces (SVIs).
- **Access Layer:** Three Layer 2 switches connect end-user devices, segregated by department, so that each department's traffic remains on its own VLAN before reaching the core.
- **End Devices:** Workstations, IP phones (Voice VLAN, Management/Admin), and a Wireless Access Point (Creative/Design and meeting rooms) connect at the access layer.

2.2 Physical topology diagram



3. Logical topology

3.1 VLAN Design

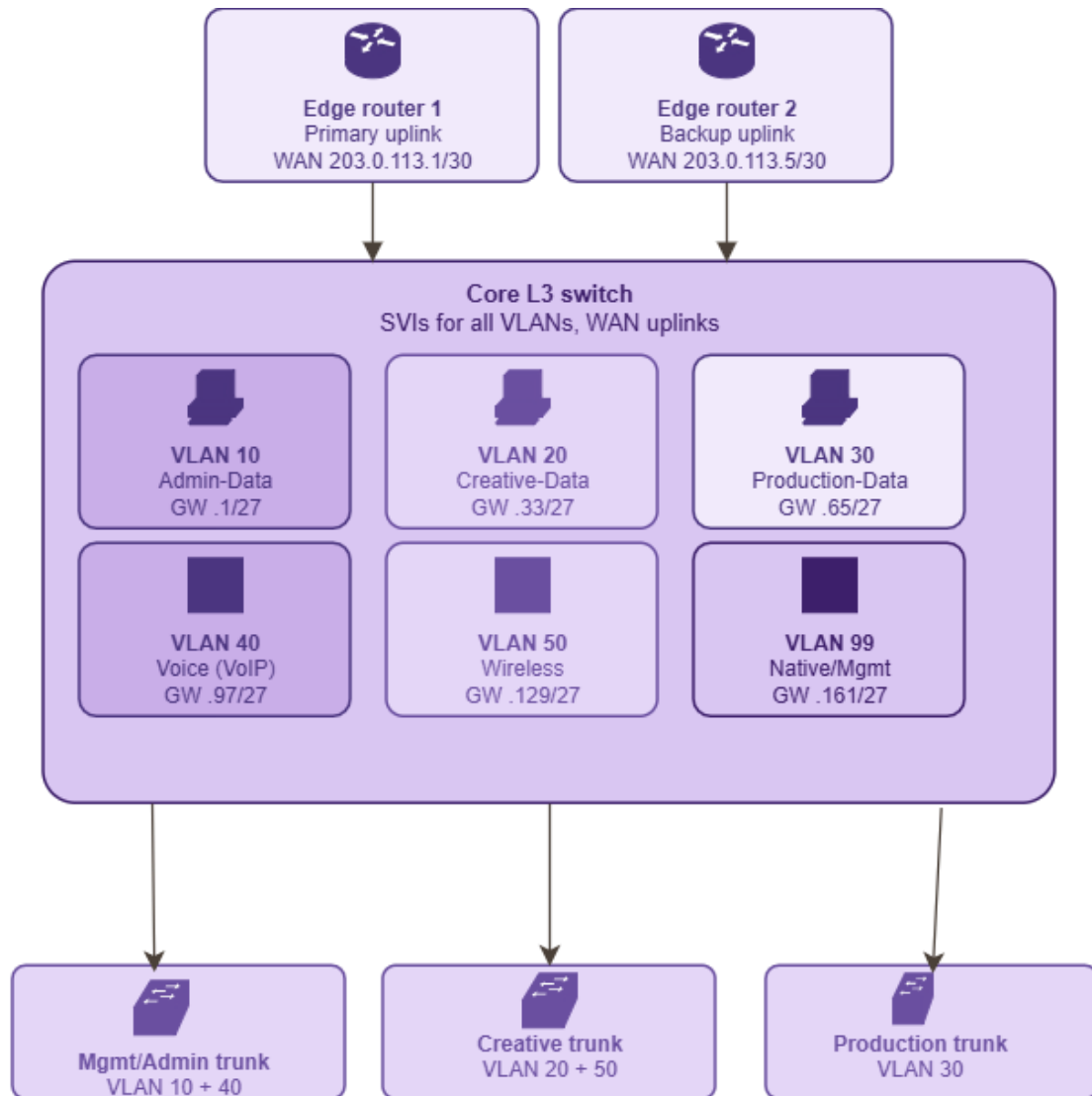
VLANs are used to segregate departmental traffic and to satisfy the design constraint isolating VoIP traffic from data traffic. The core Layer 3 multilayer switch performs inter-VLAN routing between all departments using SVIs.

VLAN ID	VLAN Name	Purpose / Traffic Type	Assigned Department
10	Admin-Data	Management and administration PCs (data)	Management/Admin
20	Creative-Data	High-bandwidth design and rendering workstations (data)	Creative/Design
30	Production Data	Logistics and operations PCs (data)	Production/Logistics
40	Voice (VoIP)	Dedicated IP phone traffic (fulfils the design constraint)	Management/Admin
50	Wireless	WAP management and client traffic (WPA2-PSK)	Creative/Design & Meeting Rooms
99	Native/Mgmt	Native VLAN for trunking and switch management	Network Infrastructure

3.2 Logical Segmentation Strategy

- **Data Segregation:** VLANs 10, 20, and 30 keep departmental data traffic separate, reducing unnecessary broadcast traffic between departments and allowing department-specific security policies.
- **Voice Separation (Constraint):** VLAN 40 is dedicated exclusively to IP phones in Management/Admin. This isolates voice traffic from office PC data traffic, supports future QoS prioritisation, and directly fulfils the mandatory design constraint.
- **Wireless Isolation:** VLAN 50 is dedicated to the Wireless Access Point, separating wireless clients from the wired network. This isolates the WPA2-PSK security domain and simplifies management of wireless clients the assigned networking challenge will be configured and demonstrated on this VLAN.
- **Core Routing:** The Layer 3 core switch, using SVIs, acts as the gateway for every VLAN and routes between departments where required.

3.3 Logical topology diagram



3.4 IP addressing plan

The assigned block 192.168.56.0/24 is subnetted into /27 blocks (30 usable hosts each), providing sufficient capacity for each VLAN, with two blocks reserved for future growth.

VLAN	Name	Subnet	Usable Range	Gateway (SVI)
10	Admin-Data	192.168.56.0/27	.1 – .30	192.168.56.1
20	Creative-Data	192.168.56.32/27	.33 – .62	192.168.56.33
30	Production-Data	192.168.56.64/27	.65 – .94	192.168.56.65
40	Voice (VoIP)	192.168.56.96/27	.97 – .126	192.168.56.97
50	Wireless	192.168.56.128/27	.129 – .158	192.168.56.129
99	Native/Mgmt	192.168.56.160/27	.161 – .190	192.168.56.161
-	Reserved	192.168.56.192/27, .224/27	-	Future Expansion

WAN Links

WAN-facing links use addressing outside the assigned /24 block, as ISP handoffs are provisioned by the ISP rather than drawn from the customer's internal block.

- **Edge Router 1 ↔ ISP 1:** 203.0.113.0/30 (router-side address: .1)
- **Edge Router 2 ↔ ISP 2:** 203.0.113.4/30 (router-side address: .5)

Trunk Configuration per Access Switch

Access Switch	VLANs Carried	Native VLAN
Mgmt/Admin Switch	10 (data), 40 (voice)	99
Creative Switch	20 (data), 50 (wireless)	99
Production Switch	30 (data)	99